

QUBO Formulation: Cryptanalysis Prime Factorization Toy Problem

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Problem Statement. Given a target composite integer N and a set of M candidate factor pairs $\{(p_i, q_i)\}_{i=0}^{M-1}$, the goal is to select exactly one candidate pair such that the product of its factors equals N . Each candidate pair i is associated with a product error term $(p_i q_i - N)^2$ and an optional instance-specific penalty pen_i . The objective is to minimize the total score, defined as the product error plus the penalty, subject to the constraint that exactly one pair is selected.

Decision Variables. Let $x_i \in \{0, 1\}$ for $i \in \{0, \dots, M-1\}$ be binary decision variables. The variable x_i indicates the selection of candidate pair i :

$$x_i = \begin{cases} 1 & \text{if candidate pair } i \text{ is selected,} \\ 0 & \text{otherwise.} \end{cases}$$

QUBO Derivation. We derive the Quadratic Unconstrained Binary Optimization (QUBO) formulation by encoding the objective function and the selection constraint into a single energy function $H(\mathbf{x})$.

Step 1: Objective Function. Define the error score for candidate pair i as $E_i = (p_i q_i - N)^2 + \text{pen}_i$. The objective is to minimize the weighted sum of these scores:

$$\min_{\mathbf{x}} \sum_{i=0}^{M-1} E_i x_i.$$

Step 2: Constraint. The requirement to select exactly one pair is expressed as the equality constraint:

$$\sum_{i=0}^{M-1} x_i = 1.$$

Step 3: Penalty Term Expansion. We enforce the constraint using a quadratic

penalty term with weight $\lambda > 0$:

$$P(\mathbf{x}) = \lambda \left(\sum_{i=0}^{M-1} x_i - 1 \right)^2.$$

Expanding the square:

$$\left(\sum_{i=0}^{M-1} x_i - 1 \right)^2 = \sum_{i=0}^{M-1} x_i^2 + \sum_{i \neq j} x_i x_j - 2 \sum_{i=0}^{M-1} x_i + 1.$$

Using the idempotent property of binary variables, $x_i^2 = x_i$, and noting that $\sum_{i \neq j} x_i x_j = 2 \sum_{0 \leq i < j \leq M-1} x_i x_j$, we obtain:

$$\begin{aligned} \left(\sum_{i=0}^{M-1} x_i - 1 \right)^2 &= \sum_{i=0}^{M-1} x_i + 2 \sum_{0 \leq i < j \leq M-1} x_i x_j - 2 \sum_{i=0}^{M-1} x_i + 1 \\ &= - \sum_{i=0}^{M-1} x_i + 2 \sum_{0 \leq i < j \leq M-1} x_i x_j + 1. \end{aligned}$$

Thus, the penalty contribution is:

$$P(\mathbf{x}) = \lambda \left(- \sum_{i=0}^{M-1} x_i + 2 \sum_{0 \leq i < j \leq M-1} x_i x_j + 1 \right).$$

Step 4: Combined Hamiltonian. The total QUBO Hamiltonian $H(\mathbf{x})$ combines the objective and penalty terms:

$$H(\mathbf{x}) = \sum_{i=0}^{M-1} E_i x_i + \lambda \left(- \sum_{i=0}^{M-1} x_i + 2 \sum_{0 \leq i < j \leq M-1} x_i x_j + 1 \right).$$

Grouping terms by variable products:

$$H(\mathbf{x}) = \sum_{i=0}^{M-1} (E_i - \lambda) x_i + \sum_{0 \leq i < j \leq M-1} (2\lambda) x_i x_j + \lambda.$$

Step 5: Q Matrix and Offset. The QUBO is defined by the upper-triangular matrix Q and scalar offset c such that $H(\mathbf{x}) = \mathbf{x}^\top Q \mathbf{x} + c$. Reading off the coefficients from the general form above:

$$Q_{i,i} = E_i - \lambda, \quad \text{for } i \in \{0, \dots, M-1\}, \quad (1)$$

$$Q_{i,j} = 2\lambda, \quad \text{for } 0 \leq i < j \leq M-1, \quad (2)$$

$$c = \lambda. \quad (3)$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any infeasible solution has strictly higher energy than any feasible solution. The minimum penalty incurred by any infeasible assignment is λ (achieved when $\sum x_i = 0$ or $\sum x_i \geq 2$). The maximum possible objective value for any feasible solution is $\max_i E_i$. Therefore, choosing

$$\lambda > \max_{i \in \{0, \dots, M-1\}} E_i$$

guarantees that the energy of any infeasible solution exceeds the energy of any feasible solution, as the penalty term alone dominates the maximum possible reduction in the objective function.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0, 1\}^M$ recovers the optimal factor pair. For any feasible solution satisfying $\sum_i x_i = 1$, the penalty term vanishes, and $H(\mathbf{x})$ equals the objective value $\sum_i E_i x_i$. For any infeasible solution, the penalty term is at least λ . By the choice $\lambda > \max_i E_i$, the energy of any infeasible solution is strictly greater than the energy of any feasible solution. Consequently, the global minimum of $H(\mathbf{x})$ must be feasible and corresponds to the candidate pair with the minimum score E_i .

Illustrative Example. Consider an instance with $M = 3$ candidate pairs and target $N = 15$. Let the pairs and penalties be:

- Pair 0: $(p_0, q_0) = (3, 5)$, $\text{pen}_0 = 0$.
- Pair 1: $(p_1, q_1) = (1, 15)$, $\text{pen}_1 = 10$.
- Pair 2: $(p_2, q_2) = (5, 3)$, $\text{pen}_2 = 0$.

The error scores are $E_0 = (15 - 15)^2 + 0 = 0$, $E_1 = (15 - 15)^2 + 10 = 10$, and $E_2 = (15 - 15)^2 + 0 = 0$. We choose $\lambda = 11$, satisfying $\lambda > \max\{0, 10, 0\}$.

The Q matrix entries and offset are:

$$Q_{0,0} = 0 - 11 = -11, \quad Q_{1,1} = 10 - 11 = -1, \quad Q_{2,2} = 0 - 11 = -11, \quad (4)$$

$$Q_{0,1} = 2(11) = 22, \quad Q_{0,2} = 22, \quad Q_{1,2} = 22, \quad (5)$$

$$c = 11. \quad (6)$$

The resulting Hamiltonian is:

$$H(\mathbf{x}) = -11x_0 - x_1 - 11x_2 + 22x_0x_1 + 22x_0x_2 + 22x_1x_2 + 11.$$

Evaluating H for feasible solutions:

- $\mathbf{x} = (1, 0, 0)$: $H = -11 + 11 = 0$.
- $\mathbf{x} = (0, 1, 0)$: $H = -1 + 11 = 10$.
- $\mathbf{x} = (0, 0, 1)$: $H = -11 + 11 = 0$.

The minimum energy is 0, achieved by $\mathbf{x} = (1, 0, 0)$ or $\mathbf{x} = (0, 0, 1)$, correctly identifying pairs 0 and 2 as optimal factorizations of 15.